

A Single-Phase Grid-Connected Inverter using Phase Control Method

Nasim Ahmed

*Institute of Information and Communication Technology
Bangladesh University of Engineering and Technology
Dhaka-1000, Bangladesh
nasimahmed@pg.iict.buet.ac.bd*

Md. Ziaur Rahman Khan

*Electrical and Electronic Engineering
Bangladesh University of Engineering and Technology
Dhaka 1000, Bangladesh.
zrkhan@eee.buet.ac.bd*

Abstract --- The design of a single-phase grid-connected inverter (GCI) using the phase-control technique is presented here. The circuit has fewer harmonics and a simpler design than traditional GCI technology. The performance of GCI has a direct influence on the entire distributed generation system. The control architecture of GCI must ensure the capability of matching inverter output voltage and frequency with continuously fluctuating grid voltage and frequency. The dynamic sinusoidal pulse width modulation technique (SPWM) has been used to control the power switches of the inverter. The finished design is simulated in MATLAB/SIMULINK to ensure output results.

Keywords – single-phase inverter, GCI, Phase-control technique, Phase-Locked-Loop (PLL), H-bridge, SPWM.

I. INTRODUCTION

There are three main functions of the GCI such as control the power flow between grid and inverter, providing power with low harmonics to the grid, and implementing grid synchronization. The most essential feature of the grid for the incorporation of renewable resources is inverters. To convert direct current (DC) to alternating current (AC), inverters play the most pivotal role. Typically, electrical energy is preserved as DC in the energy storage device. This DC energy should be converted into AC before injecting it into the utility grid. The GCI varies to just a few degrees from the traditional inverter. GCIs constantly retain grid synchronization as well as regulation of power flow into the grid. The inverter is said to be synchronized with the grid when it produces equal, in-phase grid voltage and frequency. Many works have been reported on three-phase grid-connected inverters [1] while few focuses on single-phase inverters. Some of these works have been aimed at the controller design [2-4], others on the filters [5-6], and Phase-Locked-Loop (PLL) [7-8]. Various control mechanisms have been suggested in the literature to maintain better power exchange with the grid while reducing the harmonic components in the inverter output voltage and current. Among these strategies, current hysteresis control (CHC) [9-12], voltage-oriented control (VOC) [13-17] and proportional-resonant (PR)-based control [18] are most popular. The CHC is very easy to deploy and it uses a hysteresis band, which is followed by an inverter output current. Though it has many great features such as simple, robust, automatic current limiting, high dynamic response and good stability, the switching frequency of this method is variable and depends on sampling frequency and system parameters. This fluctuating

switching frequency makes it difficult to design and implement a stable filter [9-12]. The PR-based control method has major disadvantages, such as rapidly decaying response, sensitivity issues, and instability of the phase shift in current sensors.

This paper proposes a control architecture for GCI which is simple to design. It also has low harmonic contents. In Bangladesh, there are plenty of opportunities for building a solid renewable energy harvesting infrastructure. This GCI model can easily be deployed in current smart-home, peer-to-peer electricity sharing micro grid-based architecture. The proposed control algorithm is very easy to deploy in any low-cost single-chip microcontroller-based system.

II. INVERTER DESIGN

In a GCI system, a synchronization algorithm is required to integrate the inverter output voltage with the utility grid voltage. A phase-locked loop (PLL) is selected to conduct the grid synchronization. PLL, even in the case of higher-order harmonics, can easily define the phase angle of the grid voltage. The H-bridge inverter topology is deployed in this method. An LC filter is applied in this architecture. To get the accurate amplitude of the voltage, it is performed by multiplying an accurate modulation index with dynamically generated SPWM values. The block diagram of the GCI controller can be shown in Fig. 1. The proposed GCI system is modeled and simulated to verify the effectiveness of its control structure and to determine its performance parameters using MATLAB/SIMULINK libraries.

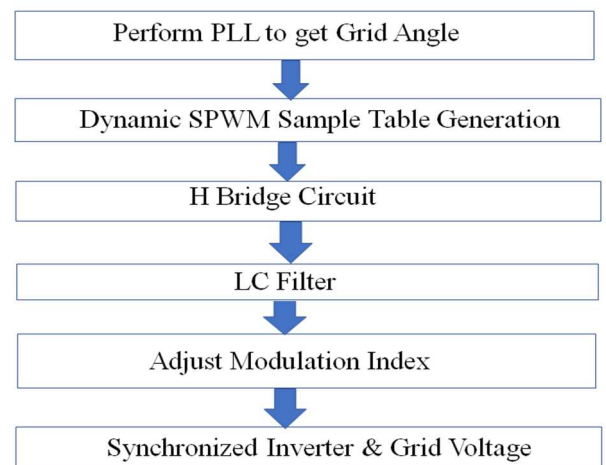


Fig. 1. Block diagram of proposed system.

A. Phase Locked Loop (PLL)

A phase-locked loop (PLL) is a control device that produces an output signal, the phase of which refers to the input signal phase. It mainly comprises a voltage-controlled oscillator (VCO) and phase detector. The oscillator produces a periodic signal, and the phase detector compares the input and VCO signal. Fig. 2 and 3 represent the PLL block diagram and its output for a certain phase of the grid voltage.

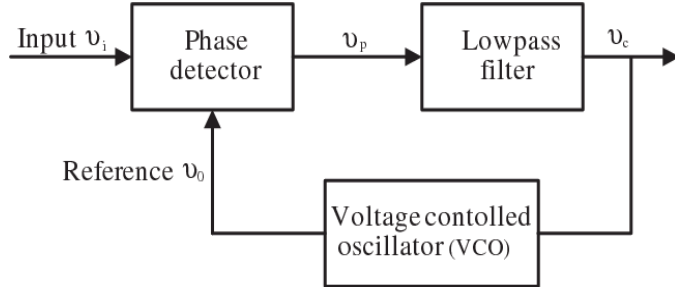


Fig. 2. Block diagram of PLL circuit.

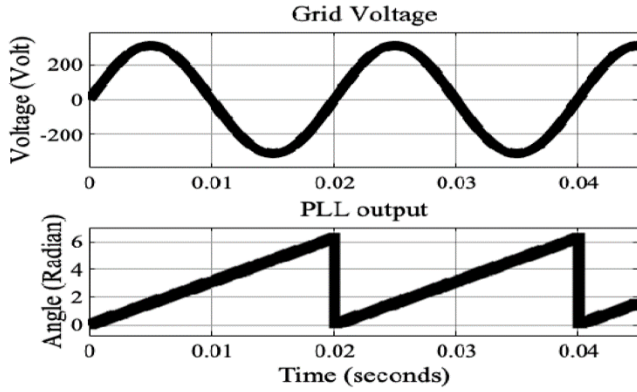


Fig. 3. PLL angle of grid voltage.

B. SPWM Generation

In the proposed sine wave inverter, an H-bridge driving configuration is used. If this H-bridge is driven with an SPWM signal, an SPWM can be found on the load. After filtering that, a pure sine wave is easily be found across the load. This is the concept of a sine wave inverter.

In terms of SPWM, the primary distinction between a stand-alone inverter and a grid-connected inverter is the lookup table. In a standalone inverter, the lookup table is static or fixed, whereas in a grid-connected inverter the PWM value comes from PLL output. After generating dynamic SPWM samples which are using for threshold value of PWM signal, are applying in gates of four switches. The proposed method uses bipolar PWM rather than unipolar PWM. In the unipolar PWM method, the output voltage fluctuates between $+V_d$ and 0 or between 0 and $-V_d$, wherein in bipolar PWM the variation occurs between $+V_d$ and $-V_d$. Fig. 4 shows the SPWM generation technique. The PWM switching frequency has been chosen 3 kHz. This reduces high harmonics distortion in output wave. There are no fixed numbers of samples for the sine table.

Though a large number of samples produce a better resolution of the pure sine wave, they also consume more memory and demand high price microcontroller.

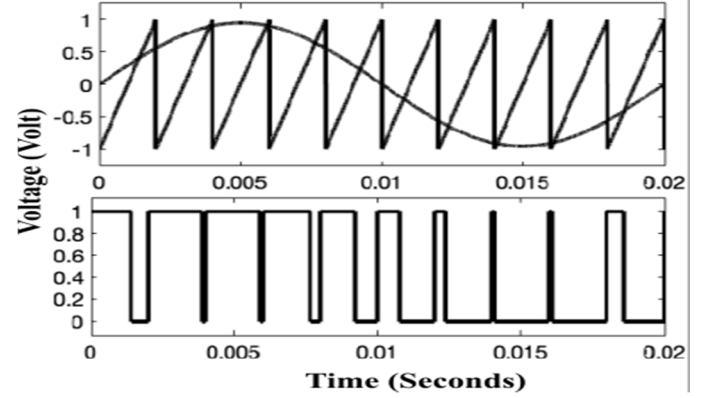


Fig. 4. SPWM Generation.

C. H Bridge

An H bridge is built with four switches. To drive the H-bridge, four signals are needed. There are two signals (S1 and S4) at 50 Hz in a push-pull configuration, and the other two (S2 and S3) are SPWM signals (3 kHz). Fig. 5 and 6 show the circuit diagram and the four signals of H-bridge circuit. The upper arms of the bridge are switched by a 50Hz signal, and the lower arms are switched by 20 kHz modulated signals. In this work, MOSFET has been used for the switching device.

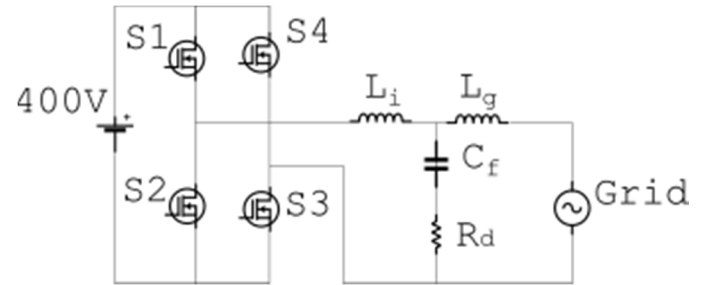
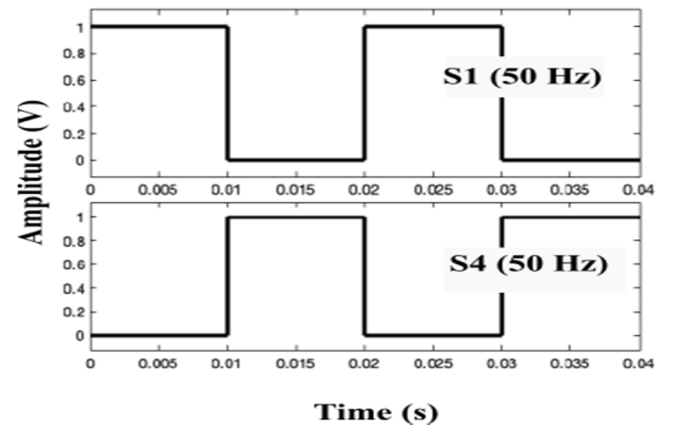


Fig. 5. Circuit diagram of the system without load.



(a)

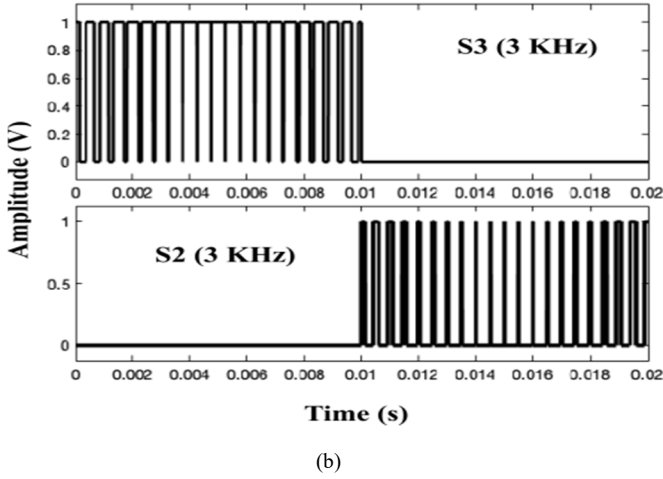


Fig. 6. Control signals of S1, S4 in (a) and S2, S3 in (b).

D. LC Filter

Before getting the final output, an LC filter has been applied in the circuit. In this experimental setup, a 10uF capacitor and 177mH inductor has been used. Before connecting to the grid, there is another inductor of 50mH has been connected by series with the inverter output terminal. Fig. 7 represents the final synchronized output after the LC filter step.

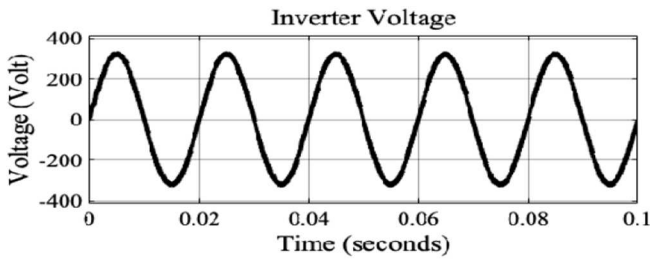
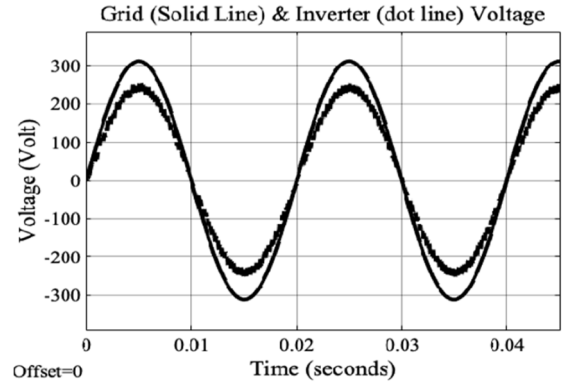


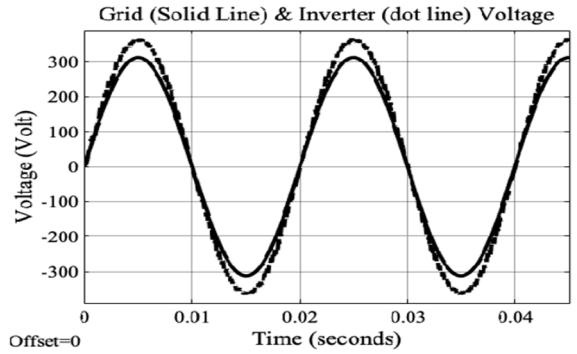
Fig. 7. Inverter voltage.

E. Adjust Modulation Index

To get the exact amplitude of the grid voltage, there must be a multiplied measured modulation index with SPWM sample values. Fig. 8 (a) represents the grid and the inverter voltage amplitude at 60% modulation index. In this condition, the grid voltage is greater than the inverter voltage. Fig. 8 (b) represents the grid and inverter voltage amplitude at a 90% modulation index. In this condition, the inverter voltage is greater than the grid voltage. In this experiment, 80% of the modulation index produces a perfectly equal inverter voltage.



(a)



(b)

Fig. 8. Grid and Inverter voltage at 60% in (a), 90% in (b) modulation index.

III. GRID CONNECTION

Unlike a current source that can be connected to the grid in parallel directly, a voltage source cannot be connected to the grid in parallel without impedance. It will create a huge circulating current between the voltage source and the grid. The designed inverter acts as a voltage source to the grid. To connect this inverter and grid, impedance is needed. In this work, an inductor has been selected, which is a non-dissipative energy storing element, will try to match the two voltage sources. Inverters with this inductor form a current source and are connected to the grid. Fig. 9 shows the designed inverter connection to the grid.

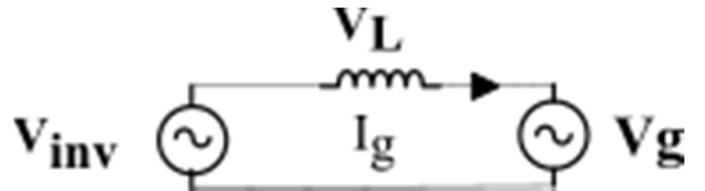


Fig. 9. Generalized Circuit Diagram.

$$V_{inv} = V_g + V_L \quad (1)$$

$$V_{inv} = V_g + L \frac{di_g}{dt} \quad (2)$$

$$V_{inv} = V_m \sin \omega t + \omega L I_m \cos \omega t \quad (3)$$

$$i_g = \frac{1}{L} \int (V_{inv} - V_g) dt \quad (4)$$

$$i_g = \sin \omega t \quad (5)$$

There are many control methods to regulate the inverter output current injection to the utility grid. The proposed system uses the phase-control method to inject inverter current to the grid. From equations [1-5], it is shown that both the grid voltage and current are in phase with inverter voltage. The proposed system takes the leverage of the popular power system formula, which has been shown in the system parameters context in equations 6 and 7.

$$P = \frac{V_{inv} V_g}{\omega L} \sin \delta \quad (6)$$

$$Q = \frac{(V_{inv} - V_g) V_g}{\omega L} \quad (7)$$

To deliver active current to the grid, the inverter needs to generate a voltage same as grid voltage, but it has to lead the grid voltage by some angle. By adjusting this angle, this active current can be increased or decreased. Fig. 10 shows the phasor diagram of active current transfer to the grid. Both the V_g and V_{inv} voltages are almost same but V_{inv} leads V_g by angle δ .

To deliver lagging reactive current to the grid, the inverter needs to generate a higher voltage than grid voltage but both are aligned in phase.

To deliver leading reactive current to the grid, the inverter needs to generate a lower voltage than grid voltage but both are aligned in phase.

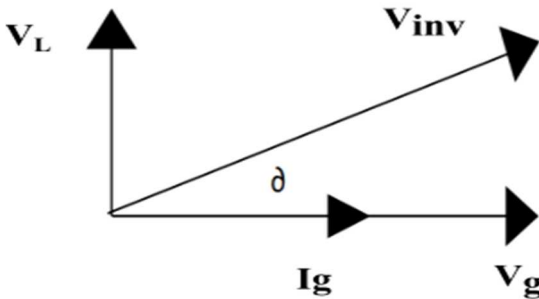
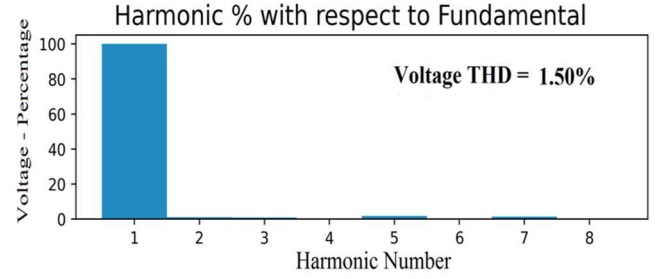


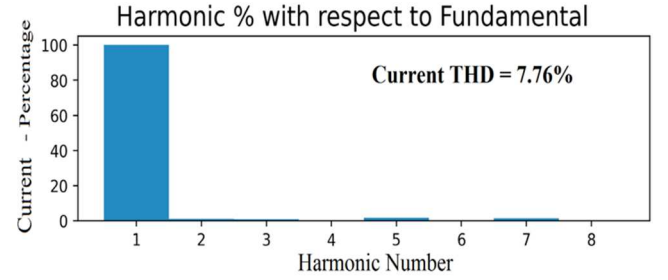
Fig. 10. Phase Diagram of Active Current Transfer.

IV. RESULT

In this experiment, the voltage THD and current THD of inverter are measured as 1.50% and 7.76%. Fig. 11 shows the harmonics of system parameters. The Inverter model uses the SPWM scheme, which is a widely accepted technique and the switching frequency is 3 kHz. The GCI power rating is 1KVA and the DC voltage of this system is 400V. In Fig. 12, Inverter supplies its excess energy into the grid because its load demand is 5A, but it produces 10A, so it feeds 5A into the grid. The phase angle difference between the grid and the inverter current is 180 degrees. Alternatively, the inverter injects lagging reactive current to the grid.



(a)



(b)

Fig. 11. Harmonics of the output parameters of the inverter (a) voltage (b) current.

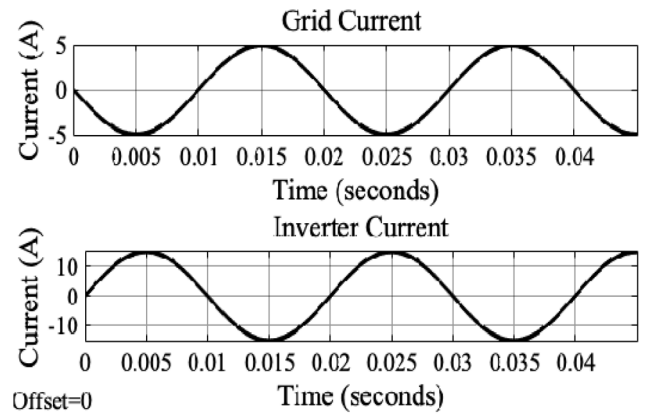


Fig. 12. Inverter Transfer Reactive Current into the Grid.

V. CONCLUSION

A single-phase grid-tied inverter has been designed, developed, and simulated in MATLAB/SIMULINK by using the phase-control technique. Due to the low harmonics distortion, and simplicity, this control method is suitable for grid-connected inverter. Modeling and simulating the suggested GCI system is to establish the validity of its control structure and to measure its output parameters to assess its performance using MATLAB/SIMULINK libraries. Finally, the MATLAB/SIMULINK observations have been demonstrated to justify the control mechanism of the implemented system. Eventually, based on the strengths of the proposed phase controller algorithm, it is established that this control scheme is the right way to execute the GCI. The suggested control method is easy to deploy in any affordable system on chip (SoC) microcontroller-based system.

ACKNOWLEDGMENT

The authors would like to acknowledge the opportunity and the infrastructural support provided by the Department of Electrical and Electronic Engineering of Bangladesh University of Engineering and Technology (BUET) to carry out this research.

REFERENCES

- [1] P. A. Pattanaik, N. K. Pilli and S. K. Singh, "Design, simulation & performance evaluation of three phase grid connected PV panel," 2015 *IEEE Power, Communication and Information Technology Conference (PCITC)*, Bhubaneswar, 2015, pp. 195-200.
- [2] S. Golestan, M. Monfared and J. M. Guerrero, "A novel control technique for single-phase grid-connected inverters," *Proceedings of the 2011 International Conference on Electrical Engineering and Informatics, Bandung*, 2011, pp. 1-6.
- [3] M. Monfared, M. Sanatkar e S. Golestan "Direct active and reactive power control of single-phase grid-tie converters" *IET Power Electronics*, vol. 5, pp. 1544-1550, 2012.
- [4] B. Crowhurst, E. El-Saadany, L. E. Chaar e L. Lamont, "Single-Phase Grid-Tie Inverter Control Using DQ Transform for Active and Reactive Load Power Compensation" 2010 *IEEE International Conference on Power and Energy*, 2010.
- [5] H. Kim and K. Kim, "Filter design for grid connected PV inverters," 2008 *IEEE International Conference on Sustainable Energy Technologies*, Singapore, 2008, pp. 1070-1075.
- [6] H. Cha and T. Vu, "Comparative analysis of low-pass output filter for single-phase grid-connected Photovoltaic inverter," 2010 *Twenty-Fifth Annual IEEE Applied Power Electronics Conference and Exposition (APEC)*, Palm Springs, CA, 2010, pp. 1659-1665.
- [7] Y. Yang and F. Blaabjerg, "Synchronization in single-phase grid-connected photovoltaic systems under grid faults," 2012 3rd *IEEE International Symposium on Power Electronics for Distributed Generation Systems (PEDG)*, Aalborg, 2012, pp. 476-482.
- [8] S. Conti, S. Raiti and G. Tina, "Simulink modelling of LV photovoltaic grid-connected distributed generation," *CIREN 2005 - 18th International Conference and Exhibition on Electricity Distribution*, Turin, Italy, 2005, pp. 1-5.
- [9] R. Ichikawa, H. Funato and K. Nemoto, "Experimental verification of single phase utility interface inverter based on digital hysteresis current controller," 2011 *International Conference on Electrical Machines and Systems*, Beijing, 2011, pp. 1-6.
- [10] P. A. Dahono, "New current controllers for single-phase full-bridge inverters," *International Conference on Power System Technology*, Singapore, 2004, pp. 1757-1762 Vol.2.
- [11] M. Azab, "A new direct power control of single phase PWM boost converter," 2003 *46th Midwest Symposium on Circuits and Systems*, Cairo, 2003, pp. 1081-1084 Vol. 3.
- [12] P. A. Dahono, "New hysteresis current controller for single-phase full-bridge inverters," in *IET Power Electronics*, vol. 2, no. 5, pp. 585-594, Sept. 2009.
- [13] B. Bahrani, A. Rufer, S. Kenzelmann and L. A. C. Lopes, "Vector Control of Single-Phase Voltage-Source Converters Based on Fictive-Axis Emulation," in *IEEE Transactions on Industry Applications*, vol. 47, no. 2, pp. 831-840, March-April 2011.
- [14] U. A. Miranda, L. G. B. Rolim and M. Aredes, "A DQ Synchronous Reference Frame Current Control for Single-Phase Converters," 2005 *IEEE 36th Power Electronics Specialists Conference*, Recife, 2005, pp. 1377-1381.
- [15] J. Gong, B. Chen, P. Li, F. Liu and X. Zha, "Feedback decoupling and distortion correction based reactive compensation control for single-phase inverter," 2009 *International Conference on Power Electronics and Drive Systems (PEDS)*, Taipei, 2009, pp. 1454-1459.
- [16] S. Samerchur, S. Premrudeepreechacharn, Y. Kumsuwun and K. Higuchi, "Power control of single-phase voltage source inverter for grid-connected photovoltaic systems," 2011 *IEEE/PES Power Systems Conference and Exposition*, Phoenix, AZ, 2011, pp. 1-6.
- [17] M. M. N. Amin and O. A. Mohammed, "Vector oriented control of voltage source PWM inverter as a dynamic VAR compensator for wind energy conversion system connected to utility grid," 2010 *Twenty-Fifth Annual IEEE Applied Power Electronics Conference and Exposition (APEC)*, Palm Springs, CA, 2010, pp. 1640-1650.
- [18] M. Ciobotaru, R. Teodorescu, F. Blaabjerg, "Control of single-stage single-phase PV inverter". *Proc. IEEE 11th Eur. Conf. on Power Electronics Application*, Dresden, Germany, September 2005, p. 10.
- [19] M. Tina and G. Celsa, "A Matlab/Simulink model of a grid connected single-phase inverter," 2015 *50th International Universities Power Engineering Conference (UPEC)*, Stoke on Trent, 2015, pp. 1-6, doi: 10.1109/UPEC.2015.7339820.
- [20] M. H. Rashid, *Power Electronic: Circuits, Devices and Applications*, 4th ed. Pearson, NY, 2013.
- [21] S. Shuvo, E. Hossain and Z. Khan, 2020. "Fixed Point Implementation of Grid Tied Inverter in Digital Signal Processing Controller". *IEEE Access*, 8, pp.89215-89227.
- [22] A. Abdalrahman, A. Zekry. "Control of the grid-connected inverter using dsPIC microcontroller". 2013 *Second International Japan-Egypt Conference on Electronics, Communications and Computers (JEC-ECC)*, 2013.